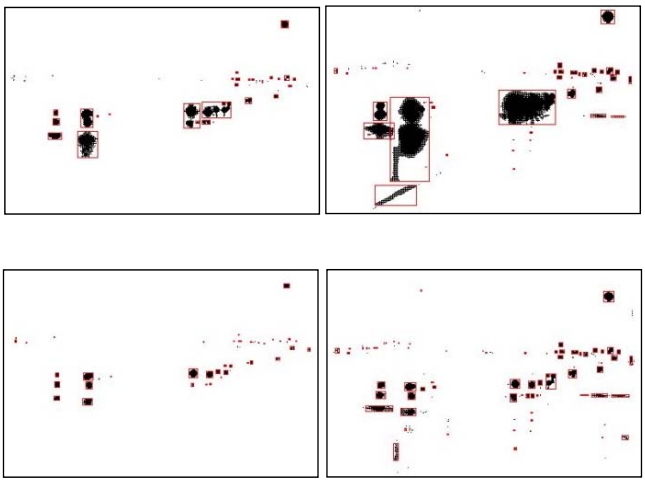
Computer Vision
Lecture 12: Binary Image Processing

Yeungnam University
MSP lab.
Ho-Youl JUNG

Binary image Labeling

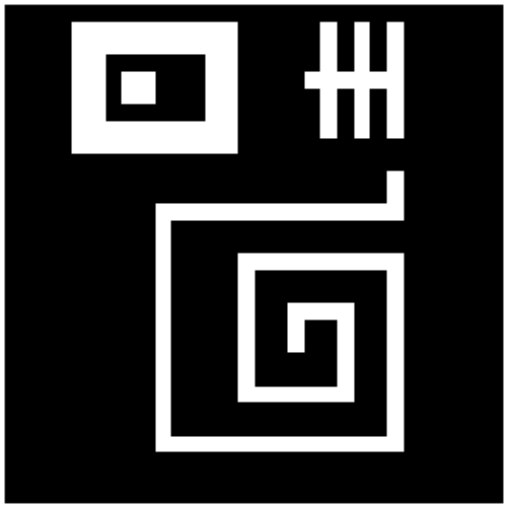
- After thresholding an image, we want to know usually the following about the regions found ...
- How many objects are in the image?
- Where are the distinct “object” components?





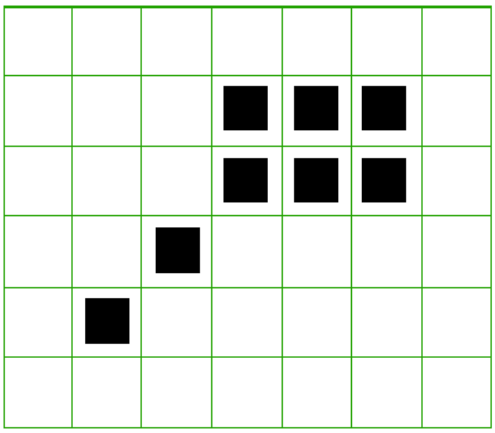
Connectivity

- How many components are there?



Connectivity

- Neighbor Metric

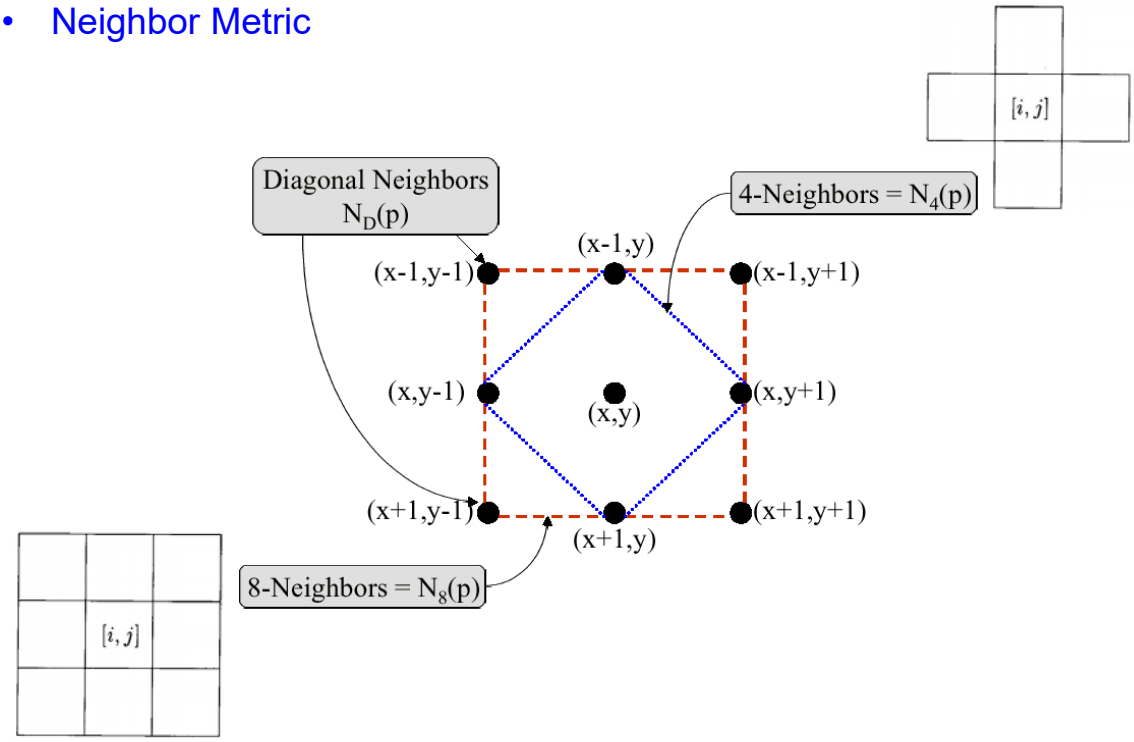


3 components ?

1 component ?

Connectivity

- Neighbor Metric



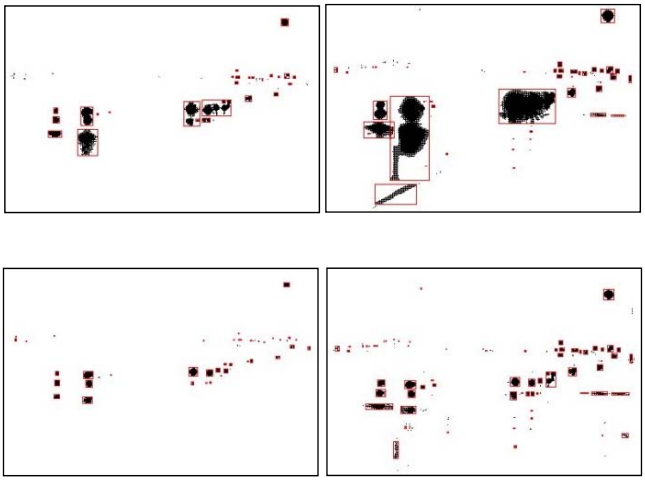
Binary Image Labeling



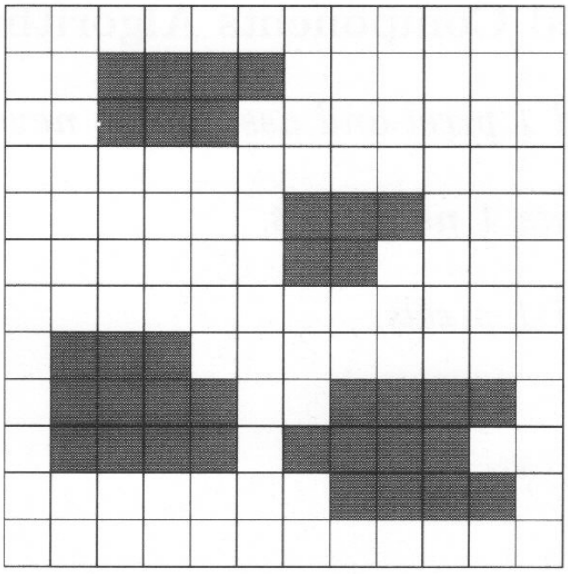
Binary Image Labeling

Global Fixed/ Global Automatic/
Local Adaptive I/II thresholding methods





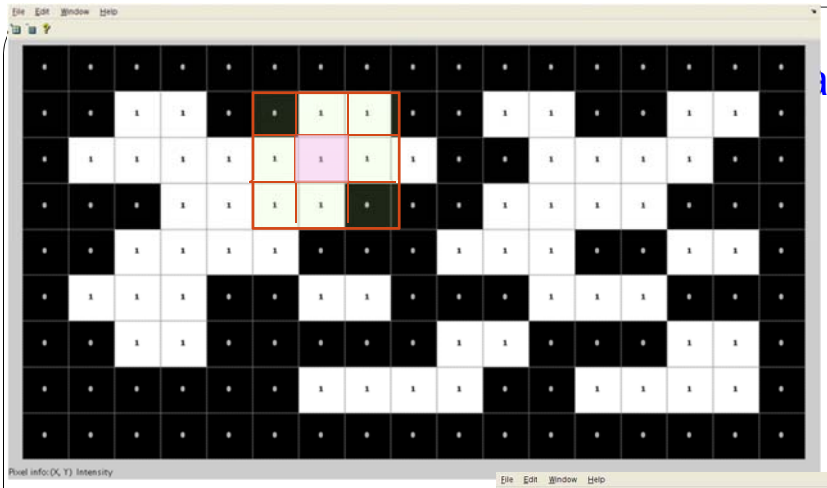
Binary Image Labeling



		1	1	1	1						
		1	1	1							
						2	2	2			
						2	2				
	3	3	3								
	3	3	3	3			4	4	4	4	
	3	3	3	3		4	4	4	4		
							4	4	4	4	

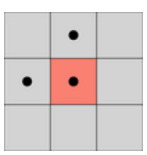
Binary Image Labeling : Two-pass scanning

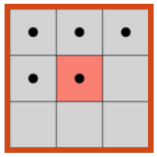
- On the first pass:
  - Iterate through each element of the data by column, then by row (Raster Scanning)
  - If the element is not the background
    - Get the neighboring elements of the current element
    - If there are no neighbors, uniquely label the current element and continue
    - Otherwise, find the neighbor with the smallest label and assign it to the current element
    - Store the equivalence between neighboring labels
- On the second pass:
  - Iterate through each element of the data by column, then by row
  - If the element is not the background
    - Relabel the element with the lowest equivalent label

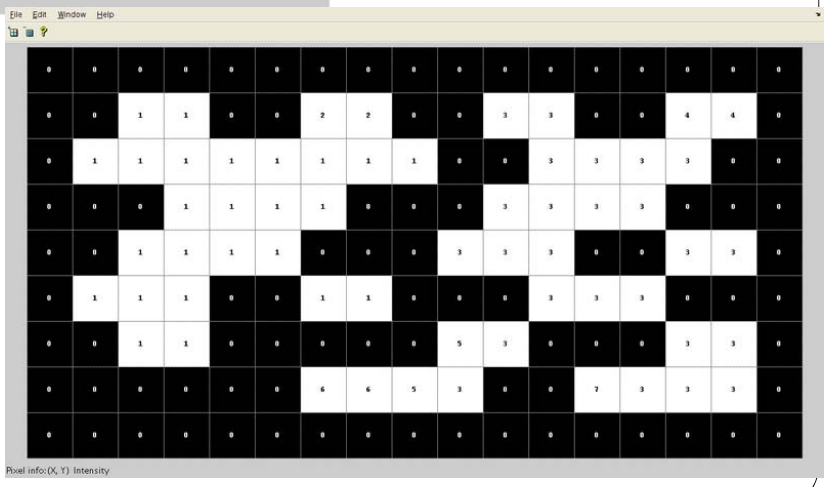


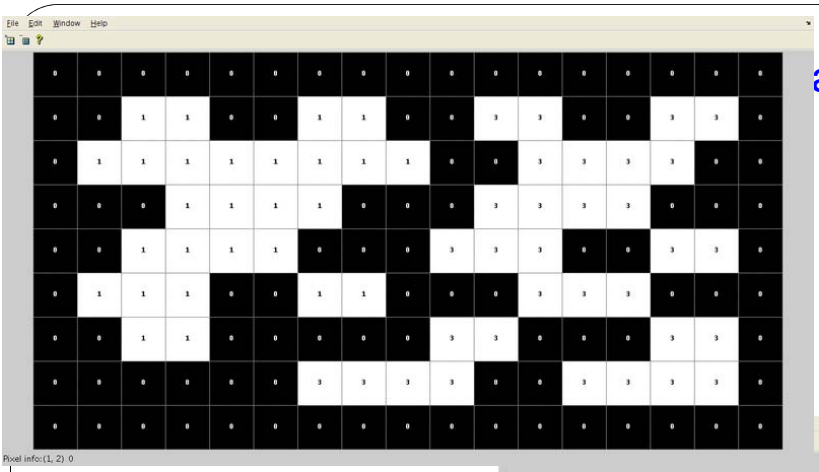
anning

Set ID	Equivalent Labels
1	1,2
2	1,2
3	3,4,5,6,7
4	3,4,5,6,7
5	3,4,5,6,7
6	3,4,5,6,7
7	3,4,5,6,7



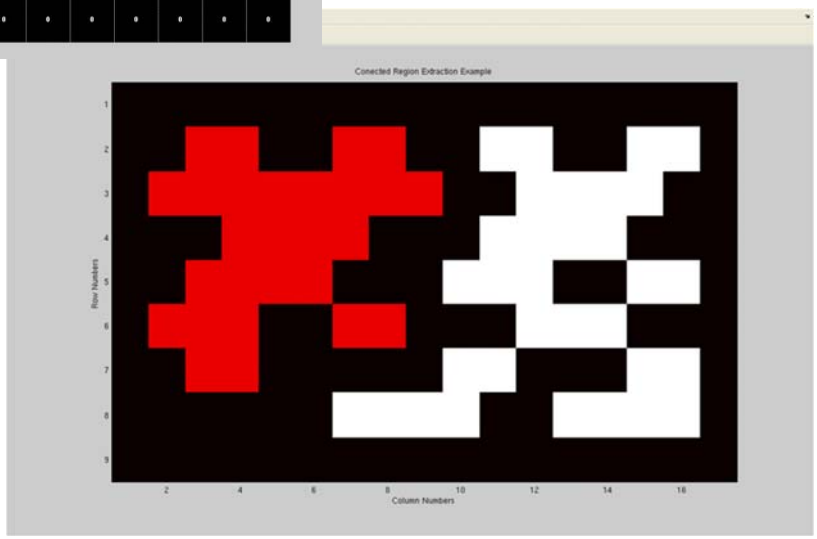






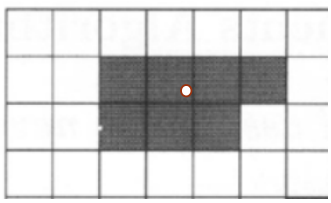
anning

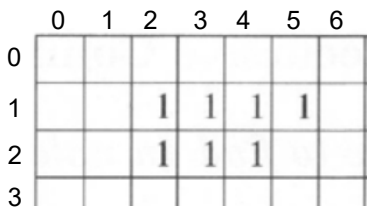
Set ID	Equivalent Labels
1	1,2
2	1,2
3	3,4,5,6,7
4	3,4,5,6,7
5	3,4,5,6,7
6	3,4,5,6,7
7	3,4,5,6,7



Position

- Position of object plays an important role in many applications
- Center of area of the object or Bounding box





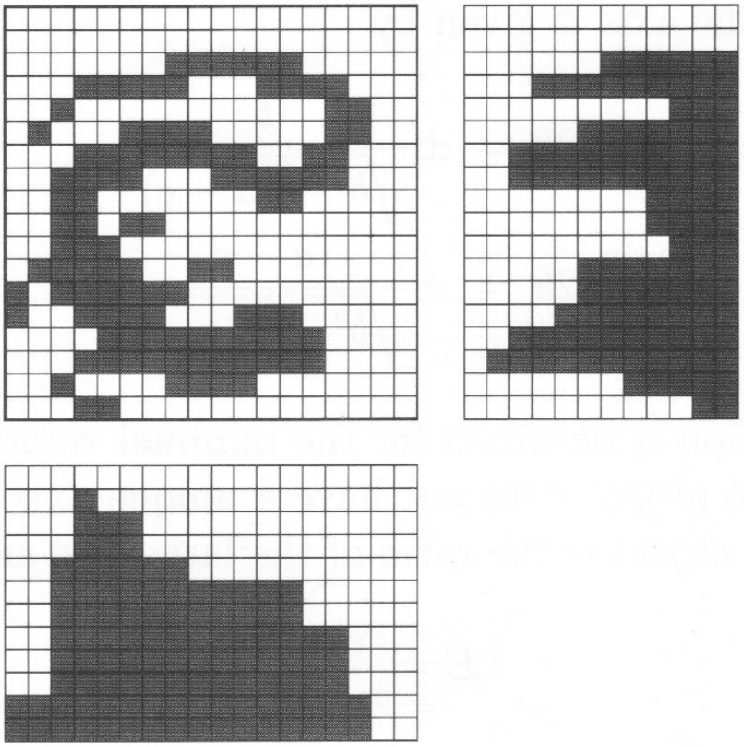
$$x_c = \frac{1}{|B|}\sum_{i=0}^{n-1}\sum_{j=0}^{m-1} iB[i,j]$$

$$y_c = \frac{1}{|B|}\sum_{i=0}^{n-1}\sum_{j=0}^{m-1} jB[i,j]$$

$$x_c = \frac{1}{7}(2+3+4+5+2+3+4) = 3.28$$

$$y_c = \frac{1}{7}(1+1+1+1+2+2+2) = 1.43$$

Binary Image Features : Projection



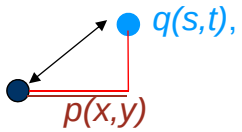
Distance Measure

- In many applications, it is necessary to find the distance between t wo pixels or two components of image
- Given *p(x,y)*, *q(s,t)*, and *z(u,v)*, *D* is a *distance function* or *metric* if

  (a) $D(p,q) > 0$, $(D(p,q) = 0 \text{ iff } p = q)$,

  (b) $D(p,q) = D(q,p)$, and

  (c) $D(p,z) < D(p,q) + D(q,z)$.



Euclidean distance

$$D_e(p,q) = \sqrt{(x-s)^2 + (y-t)^2}$$

City-block distance

$$D_4(p,q) = |x-s| + |y-t|$$

Chessboard distance

$$D_8(p,q) = \max(|x-s|, |y-t|)$$

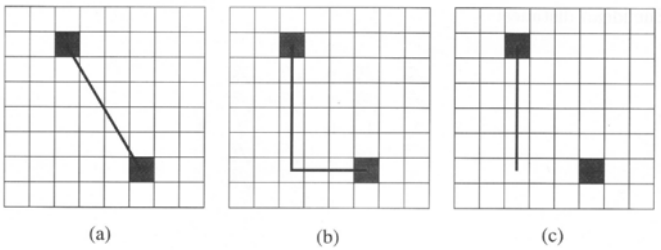
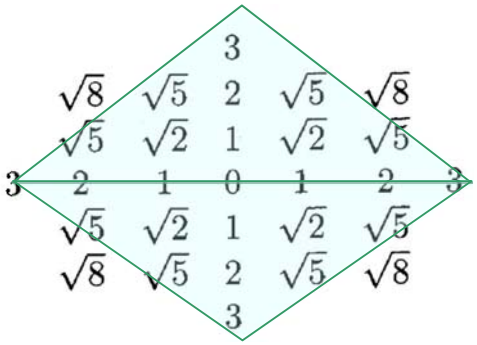
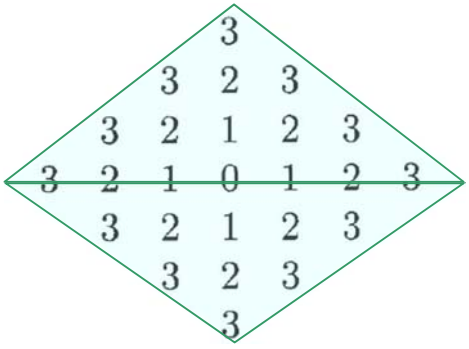


Figure 2.17: Examples of (a) Euclidean, (b) city-block, and (c) chessboard distance measures.

Distance



Euclidean distance



City Block distance

Distance Transform

- The distance transform is an operator normally only applied to binary images.
- The result of the transform is a graylevel image that looks similar to the input image, except that the graylevel intensities of points inside foreground regions are changed to show the distance to the closest boundary from each point
- The minimum distance between pixels of object and background is used

0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	0
0	1	1	1	1	1	1	0
0	1	1	1	1	1	1	0
0	1	1	1	1	1	1	0
0	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0

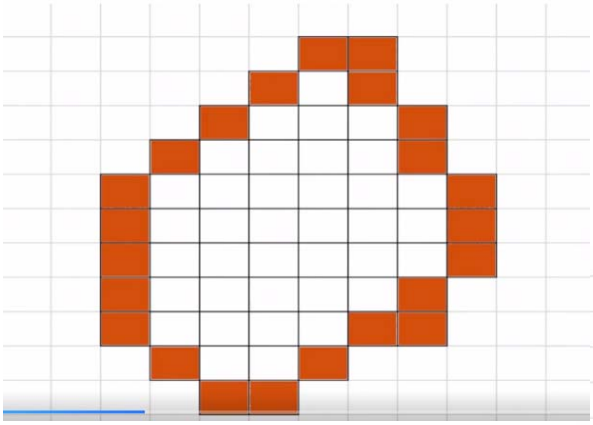
⇒

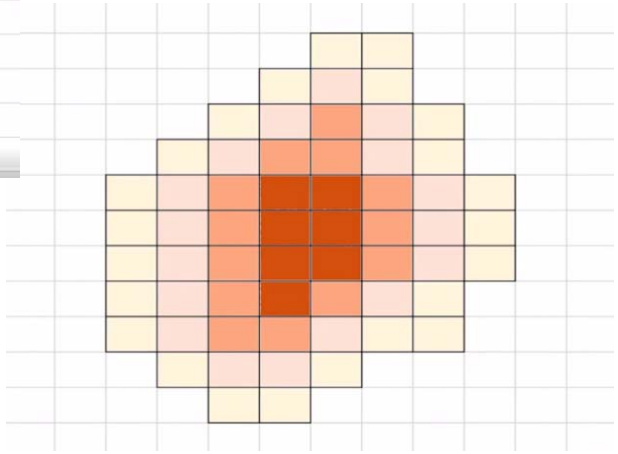
0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	0
0	1	2	2	2	2	1	0
0	1	2	3	3	2	1	0
0	1	2	2	2	2	1	0
0	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0

https://webnautes.tistory.com/1280



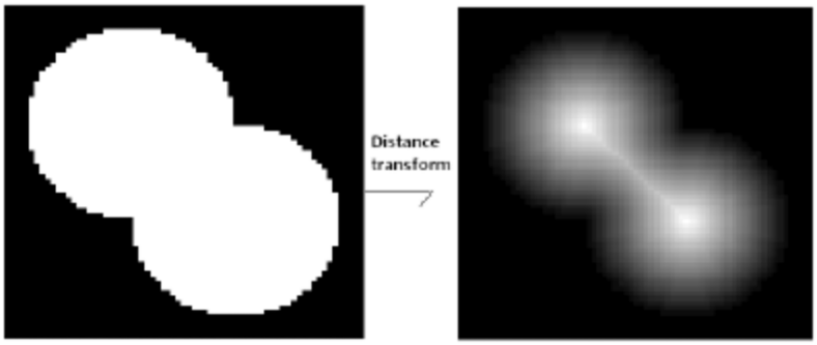
Distance Transform





Distance Transform

https://www.researchgate.net/figure/Distance-transform-of-the-binary-image-composed-with-a-white-pixel-form-and-a-black-pixel_fig2_220998362

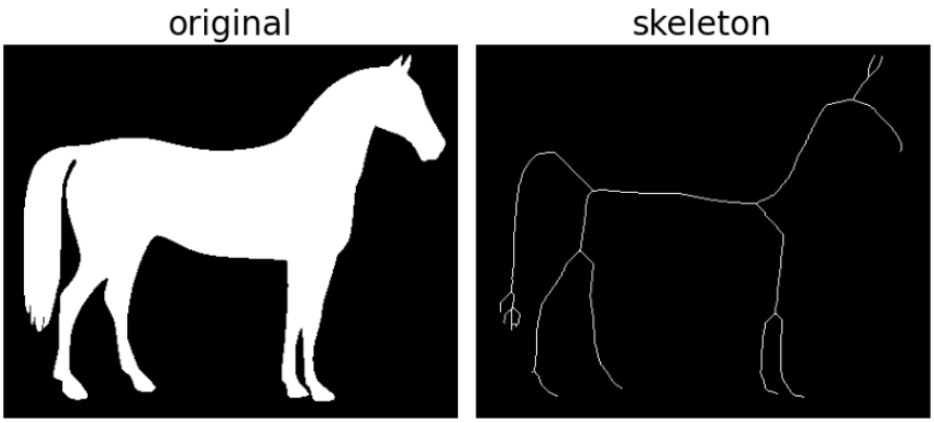




Skeletonization/Medial Axis Transform

- Process for reducing foreground regions in a binary image to a skeletal remnant that largely preserves the extent and connectivity of the original region while throwing away most of the original foreground pixels





Thank You
Next : Morphological Operation

